

Week 2 Technical Report - Divide and Conquer

Project repository (complete source, tests, benchmarks and results):

<https://github.com/RDeibel2025/CTX-Advanced-Algorithms>

Full recurrence derivations: [analysis/week2_recurrences.md](#) · runnable demonstration: [examples/week2_demo.py](#)

Robert Deibel · CSC 5300 Advanced Algorithms · Concordia University Texas · Fall 2026



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Fourteen Master Theorem solutions.

1. Executive Summary

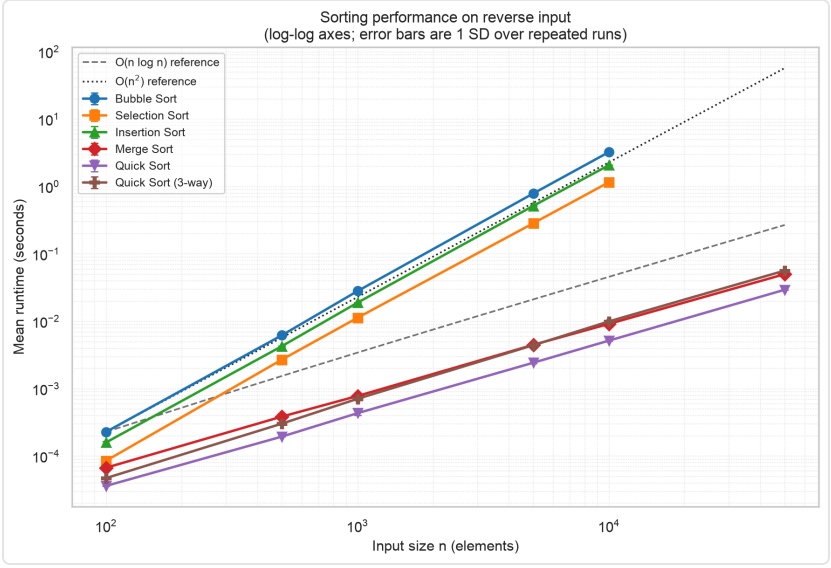
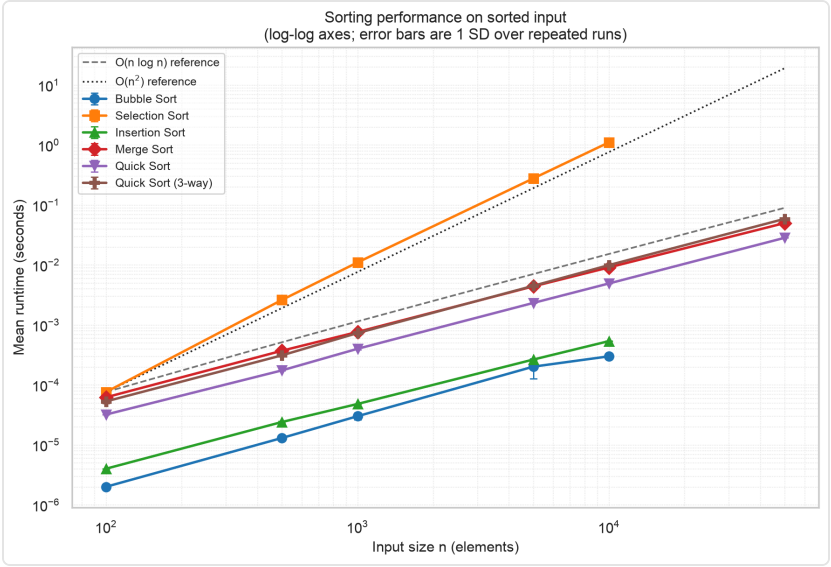
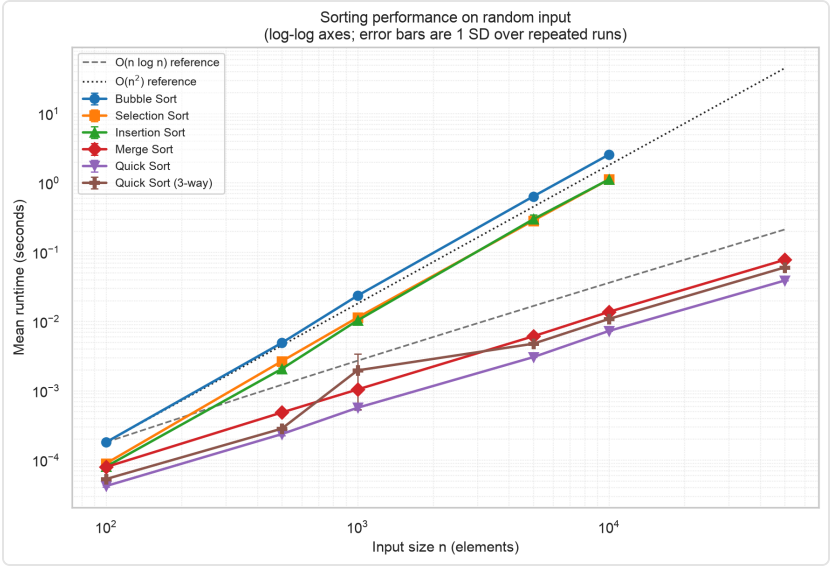
Merge sort and quicksort beat the Week 1 quadratics by a compounding factor - parity at $n = 100$, $154\times$ at $n = 10,000$ - and both matched $O(n \log n)$, with doubling ratios of 2.03-2.20 and 2.15-2.39 against a predicted ~ 2.1 . Quicksort's $O(n^2)$ worst case never appeared. The most interesting result was negative: three-way partitioning rescues duplicate-heavy input only from a weakness Hoare's scheme lacks.

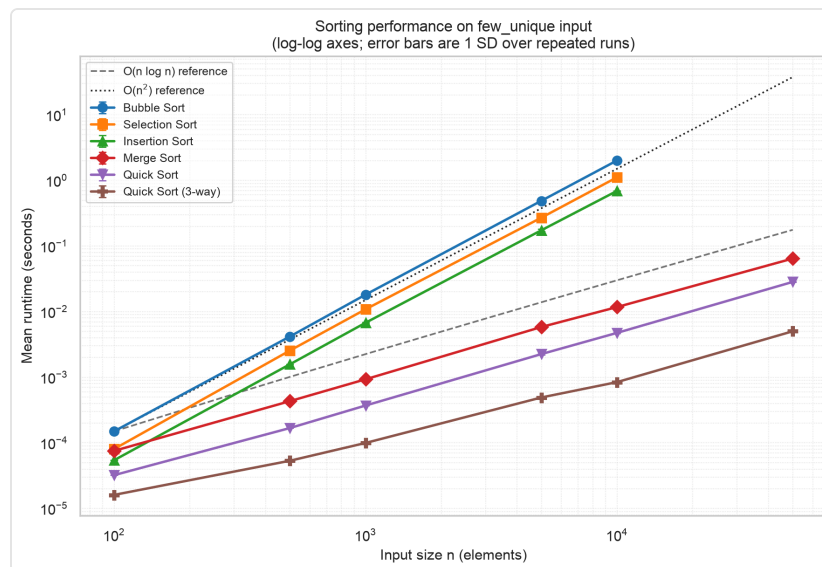
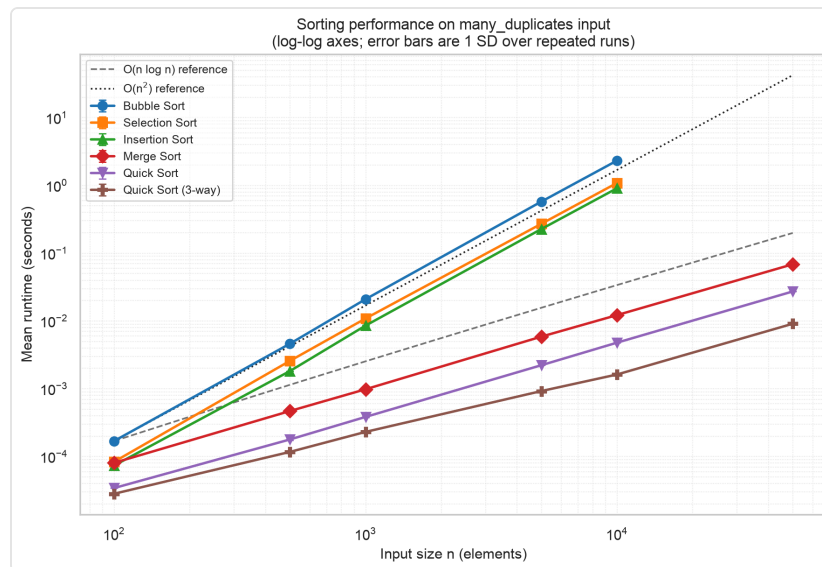
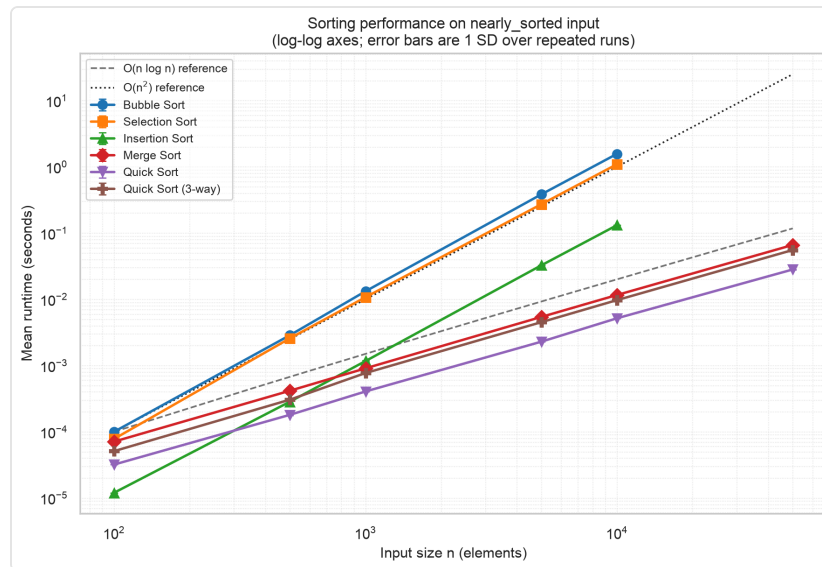
2. Methodology

Apple M2 Max, 12 cores, 64 GB RAM; macOS 14.5 (arm64); CPython 3.12.4; timing by `time.perf_counter()`. Six algorithms (three Week 1 quadratics, merge, quicksort, quicksort three-way) over sizes 100 ... 50,000 and six shapes: random, sorted, reverse, nearly sorted (5% transposed), many duplicates (10 distinct values), few unique (3). Two warm-up runs are discarded per cell, five measured below $n = 10,000$ and three above. Generation is seeded; every algorithm gets the identical list per cell, and every run is verified a sorted permutation. Median relative SD: 0.7%.

The $O(n^2)$ size cap. The quadratics were measured only to $n = 10,000$; the eighteen cells at $n = 50,000$ appear in [comparison_table.csv](#) as `omitted` with reasons. Fitting $t = c \cdot n^2$ projects 64 s, 28 s and 28 s per run on random data - about an hour. I measured them anyway and checked the projection (§4).

3. Results





Mean seconds at n = 10,000:

Algorithm	random	sorted	reverse	nearly	many dup	few uniq
Bubble	2.5498	0.0003	3.2360	1.5665	2.3059	1.9940
Selection	1.1182	1.1031	1.1556	1.0929	1.0782	1.1047
Insertion	1.1167	0.0005	2.0667	0.1313	0.9004	0.6855
Merge	0.0137	0.0091	0.0091	0.0116	0.0121	0.0116
Quick	0.0073	0.0049	0.0052	0.0052	0.0047	0.0047
Quick 3-way	0.0108	0.0099	0.0099	0.0097	0.0016	0.0008

Speedup over the best quadratic:

n	100	500	1,000	5,000	10,000
Merge	1.0×	4.3×	9.9×	46.7×	81.5×
Quick	1.9×	8.7×	18.2×	93.2×	153.6×

The gap compounds; at $n = 100$ insertion and merge sort are indistinguishable (0.000080 s vs 0.000079 s, inside the run-to-run variation). Asymptotic superiority is a claim about large n .

Merge sort was among the most consistent - 1.5× variation across the six shapes at $n = 10,000$, against 10,823× for bubble and 3,856× for insertion. Its positional split gives the same recursion tree whatever the input order: no best or worst case to find. The price is $O(n)$ auxiliary space.

QuickSort was fastest on five of six shapes, spread 1.5×: best on few-unique (0.0047 s), worst on random (0.0073 s) - nothing like the $O(n \log n)$ -to- $O(n^2)$ range theory permits.

Surprising finding. Three-way partitioning is *slower* than two-way wherever values are mostly distinct - roughly half speed - paying off only on duplicate-heavy input: 2.9× at ten distinct values, 5.6× at three.

4. Complexity Validation

Observed $t(2n)/t(n)$, mean of the 500→1,000 and 5,000→10,000 steps:

Algorithm	random	sorted	reverse	few uniq	Predicted
Bubble	4.40	1.90	4.32	4.23	4 / 2 adaptive
Selection	4.11	4.11	4.13	4.19	4
Insertion	4.38	2.01	4.22	4.14	4 / 2 adaptive
Merge	2.20	2.07	2.03	2.08	~2.1
Quick	2.39	2.20	2.18	2.15	~2.1

Merge sort stays within 0.10 of prediction everywhere. Bubble and insertion drop to ~2.1 on sorted input, their early exit appearing as a change of complexity class.

Did I observe quicksort's $O(n^2)$ worst case? No - every shape stayed at or below 2.4, including the three meant to be adversarial. Three mechanisms prevent it: randomized pivots leave no fixed input adversarial, so the worst case needs an unlucky *sequence of draws*; Hoare's partition splits a run of equal elements near its

middle rather than onto one side; and recursing into the smaller partition caps stack depth at $O(\log n)$ - 10-14 frames across the six shapes at $n = 100,000$, against Python's 1,000-frame limit.

Unreachable *by this implementation* is not unreachable. Lomuto's partition, the scheme most often taught, reaches it at once:

Scheme (3 values)	growth/doubling	t(16,000)
Lomuto (2-way)	3.98	1.9246 s
Hoare (2-way)	2.15	0.0079 s
Dutch flag (3-way)	2.02	0.0016 s

3.98 against a predicted 4 is the worst case, measured - 244× slower than Hoare on identical input.

The capped cells. Measuring all eighteen put the fitted $t = c \cdot n^2$ projection within **2.7%** on the sixteen genuinely quadratic cells (mean 1.4%), and off by **+436%** on the two sorted-input cells, where these algorithms are linear and n^2 is wrong. The extrapolation holds where its assumption does.

5. Optimization Impact

Insertion-sort cutoff, measured by varying the threshold. On random input at $n = 50,000$, disabling it costs 0.0563 s; a threshold of 10 gives 0.0389 s (**1.44×**) and the optimum of 20 gives 0.0353 s (1.59×). Past 20 the curve turns back up as insertion sort's $O(k^2)$ reasserts itself; on nearly-sorted and few-unique input it was still improving at 80, so one constant is a compromise.

Three-way partitioning: 5.6× at three distinct values, 2.9× at ten, ~2× slower otherwise. **Randomized pivot** cannot be isolated by timing; its evidence is §4's negative result.

6. Practical Recommendations

- **n below ~100:** insertion sort - it matches merge sort.
- **General purpose:** quicksort - randomized pivot, Hoare partition, insertion cutoff.
- **Stability or a guarantee:** merge sort, 1.9× slower on random.
- **Duplicate-heavy data:** three-way partitioning.
- **Never use Lomuto without a duplicate guard.**

7. Conclusion

Measurement matched theory closely enough that the interesting results were where it did not: asymptotic advantage is real but only past a measurable crossover, quicksort's worst case belongs to the partition scheme rather than to quicksort, and three-way partitioning is a trade.

8. References

- Cormen, Leiserson, Rivest, Stein. *Introduction to Algorithms*, 4th ed., Ch. 2.3-2.4, 4, 7

- Amakobe. *Advanced Computational Algorithms*, 2nd ed., Ch. 2
- Sedgewick, Wayne. *Algorithms*, 4th ed., §2.3

AI use acknowledgement

I used Anthropic's Claude (Claude Code) to draft the code, tests and first draft of this report, from a specification I wrote off the instructions and assigned reading. The measurements are my own, every figure above computed from the result CSVs rather than transcribed. The design decisions were mine - Hoare's partition over Lomuto's, recursing into the smaller partition, and measuring the capped cells rather than projecting them. Full disclosure: [docs/AI_USE.md](#).

Appendix A: Master Theorem - Recurrence Relations

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Week 2 - Divide and Conquer Implementation Project

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Fourteen recurrences solved below. Ten are straightforward applications of the Master Theorem; three are included precisely because the theorem does **not** settle them (§9, §13 and §14), since knowing where a tool stops working is part of knowing how to use it. The remaining one, §11, falls in the gap between cases 2 and 3 and needs the *extended* case 2 rather than the three-case statement. That is $10 + 3 + 1 = 14$.

Abridged for this PDF. The table below solves all fourteen recurrences in full - a , b , $f(n)$, $\log_b(a)$, the case applied, why, and the resulting bound. The four whose reasoning needs more than a table row are then discussed individually. Worked derivations for the other ten are in [analysis/week2_recurrences.md](#).

The theorem, as used here

For a recurrence of the form

$$T(n) = a \cdot T(n/b) + f(n), \text{ with } a \geq 1, b > 1$$

compare $f(n)$ against the *watershed function* $n^{\log_b a}$. That exponent is what the recursion costs on its own - it counts the leaves of the recursion tree - and the three cases ask whether the work done splitting and combining is smaller, equal, or larger than that.

Case	Condition	Result
1	$f(n) = O(n^{\log_b a - \epsilon})$ for some $\epsilon > 0$	$\Theta(n^{\log_b a})$ - leaves dominate
2	$f(n) = \Theta(n^{\log_b a} \cdot \log^k n)$, $k \geq 0$	$\Theta(n^{\log_b a} \cdot \log^{(k+1)} n)$ - balanced
3	$f(n) = \Omega(n^{\log_b a + \epsilon})$ for some $\epsilon > 0$, and $a \cdot f(n/b) \leq c \cdot f(n)$ for some $c < 1$ and large n	$\Theta(f(n))$ - the root dominates

Two requirements are easy to skip past, and they are what §11 and §13 fall foul of. Cases 1 and 3 need $f(n)$ to differ from the watershed by a *polynomial* factor - an n^ϵ gap, not merely a logarithmic one. And case 3 additionally needs the **regularity condition**, which is what rules out pathologically oscillating f .

Summary

#	Recurrence	a	b	f(n)	log_b a	Case	Why	Result
1	$2T(n/2) + \Theta(n)$	2	2	$\Theta(n)$	1	2	$f = \Theta(n^{\log_b a})$ exactly	$\Theta(n \log n)$
2	$T(n/2) + \Theta(1)$	1	2	$\Theta(1)$	0	2	$f = \Theta(n^0) = \text{watershed}$	$\Theta(\log n)$
3	$2T(n/2) + \Theta(1)$	2	2	$\Theta(1)$	1	1	$\Theta(1) = O(n^{(1-1)})$	$\Theta(n)$
4	$3T(n/2) + \Theta(n)$	3	2	$\Theta(n)$	1.585	1	$n = O(n^{(1.585-0.585)})$	$\Theta(n^{1.585})$
5	$4T(n/2) + \Theta(n)$	4	2	$\Theta(n)$	2	1	$n = O(n^{(2-1)})$	$\Theta(n^2)$
6	$7T(n/2) + \Theta(n^2)$	7	2	$\Theta(n^2)$	2.807	1	$n^2 = O(n^{(2.807-0.807)})$	$\Theta(n^{2.807})$
7	$8T(n/2) + \Theta(n^2)$	8	2	$\Theta(n^2)$	3	1	$n^2 = O(n^{(3-1)})$	$\Theta(n^3)$
8	$2T(n/2) + \Theta(n)$	2	2	$\Theta(n)$	1	2	$f = \Theta(n) = \text{watershed}$	$\Theta(n \log n)$
9	$T(n-1) + \Theta(n)$	-	-	$\Theta(n)$	-	N/A	shrinks by a constant, no b	$\Theta(n^2)$
10	$2T(n/2) + \Theta(\log n)$	2	2	$\Theta(\log n)$	1	1	$\log n = O(n^{0.5})$	$\Theta(n)$
11	$2T(n/2) + \Theta(n \log n)$	2	2	$\Theta(n \log n)$	1	2 ext.	$f = \Theta(n \cdot \log^1 n)$, so $k = 1$	$\Theta(n \log^2 n)$
12	$2T(n/2) + \Theta(n^2)$	2	2	$\Theta(n^2)$	1	3	$n^2 = \Omega(n^2)$; $2 \cdot f(n/2) = 0.5 \cdot f(n)$	$\Theta(n^2)$
13	$2T(n/2) + n/\log n$	2	2	$n/\log n$	1	N/A	gap is logarithmic, not polynomial	$\Theta(n \log \log n)$
14	$T(n-1) + T(n-2) + \Theta(1)$	-	-	$\Theta(1)$	-	N/A	subproblems unequal, not $a \cdot T(n/b)$	$\Theta(\phi^n)$

Three of the fourteen (§9, §13, §14) are outside the theorem's reach, for three different reasons: subproblems that shrink by a constant amount, a gap that is logarithmic rather than polynomial, and subproblems of unequal size.

The four the table cannot hold

Three of these fall outside the Master Theorem entirely and one needs the extended case 2. The remaining ten are routine applications, fully specified by the table above.

9. Quicksort's worst case - $T(n) = T(n-1) + \Theta(n)$

a	1
b	-
f(n)	$\Theta(n)$

The Master Theorem does not apply. The theorem requires subproblems of size n/b for a constant $b > 1$ - a *fixed fraction* of the input. Here the subproblem shrinks by a constant *amount*, so there is no b to speak of.

Solve by direct expansion instead:

$$T(n) = n + (n-1) + (n-2) + \dots + 1 = n(n+1)/2$$

$$T(n) = \Theta(n^2)$$

This is the recurrence that arises when every partition puts all remaining elements on one side. Section 4 of the report records that it was never observed for the production quicksort across any of the six data shapes - but the Lomuto partition study reached it exactly, with a measured growth ratio of 3.98 per doubling against the predicted 4.

11. Case 2 extended - $T(n) = 2T(n/2) + \Theta(n \log n)$

a	2
b	2
f(n)	$\Theta(n \log n)$
log_b a	1
watershed	n

This one needs care. $f(n) = \Theta(n \log n)$ is **larger** than the watershed n , which looks like case 3 - but case 3 requires $f(n) = \Omega(n^{1+\epsilon})$ for some $\epsilon > 0$, and $n \log n$ is *not* polynomially larger than n . It grows faster by a logarithmic factor only, and $\log n = o(n^\epsilon)$ for every $\epsilon > 0$. So the **three-case Master Theorem as usually stated does not cover this recurrence**; it falls in the gap between cases 2 and 3.

The *extended* case 2 does cover it. In the form $f(n) = \Theta(n^{\log_b a} \cdot \log^k n)$ with $k = 1$:

$$T(n) = \Theta(n^{\log_b a} \cdot \log^{(k+1)} n) = \Theta(n \log^2 n)$$

Confirmed by the recursion tree: $\log_2 n$ levels, and level i does $2^i \cdot (n/2^i) \cdot \log(n/2^i) = n \cdot \log(n/2^i)$ work. Summing over $i = 0 \dots \log n - 1$ gives $n \cdot \sum(\log n - i) = n \cdot \Theta(\log^2 n)$.

The lesson is that "f is bigger than the watershed" is not enough for case 3. The gap has to be polynomial.

13. Where the Master Theorem fails - $T(n) = 2T(n/2) + n/\log n$

a	2
b	2
f(n)	$n / \log n$
log_b a	1
watershed	n

$f(n) = n/\log n$ is **smaller** than n , which suggests case 1. But case 1 requires $f(n) = O(n^{1-\epsilon})$ for some fixed $\epsilon > 0$, and that fails: for any $\epsilon > 0$,

$$\lim_{n \rightarrow \infty} [n/\log n] / n^{1-\epsilon} = \lim n^\epsilon / \log n = \infty$$

so $n/\log n$ is not $O(n^{1-\epsilon})$ for any ϵ . It is smaller than the watershed by a logarithmic factor only, never a polynomial one - the mirror image of the §11 problem. Case 2 does not apply either, since $n/\log n = n \cdot \log^{-1} n$ requires $k = -1$ and the extended case 2 requires $k \geq 0$.

The recurrence falls between cases 1 and 2, and the Master Theorem returns no answer at all.

It is still solvable, just not with this tool. By the recursion tree: there are $\log_2 n$ levels, and level i does $2^i \cdot (n/2^i)/\log(n/2^i) = n / (\log n - i)$ work. Summing over $i = 0 \dots \log n - 1$:

$$T(n) = n \cdot \sum_{i=0}^{\log n - 1} 1/(\log n - i) = n \cdot (1 + 1/2 + \dots + 1/\log n) = n \cdot H(\log n)$$

where H is the harmonic series, $H(m) = \Theta(\log m)$. Therefore:

$$T(n) = \Theta(n \log \log n)$$

A bound strictly between the $\Theta(n)$ of case 1 and the $\Theta(n \log n)$ of case 2 - which is exactly why no case could produce it. The Akra-Bazzi method handles this class of recurrence in general.

14. Fibonacci by naive recursion - $T(n) = T(n-1) + T(n-2) + \Theta(1)$

a	2 subproblems, of <i>different</i> sizes
b	-

The Master Theorem does not apply, for two independent reasons: the subproblems shrink by a constant amount rather than a constant factor (as in §9), and they are not the same size as each other, which the theorem's $a \cdot T(n/b)$ form cannot express.

The recurrence is its own solution. $T(n)$ exceeds the Fibonacci numbers themselves, which grow as $\phi^n/\sqrt{5}$ with $\phi = (1+\sqrt{5})/2$:

$$T(n) = \Theta(\phi^n) \approx \Theta(1.618^n)$$

Included as the reminder that "divide and conquer" is not automatically efficient. Splitting a problem in two only helps when the pieces are *fractions* of the original; splitting into pieces of size $n-1$ and $n-2$ duplicates almost all the work, which is what memoisation eliminates to give $\Theta(n)$.